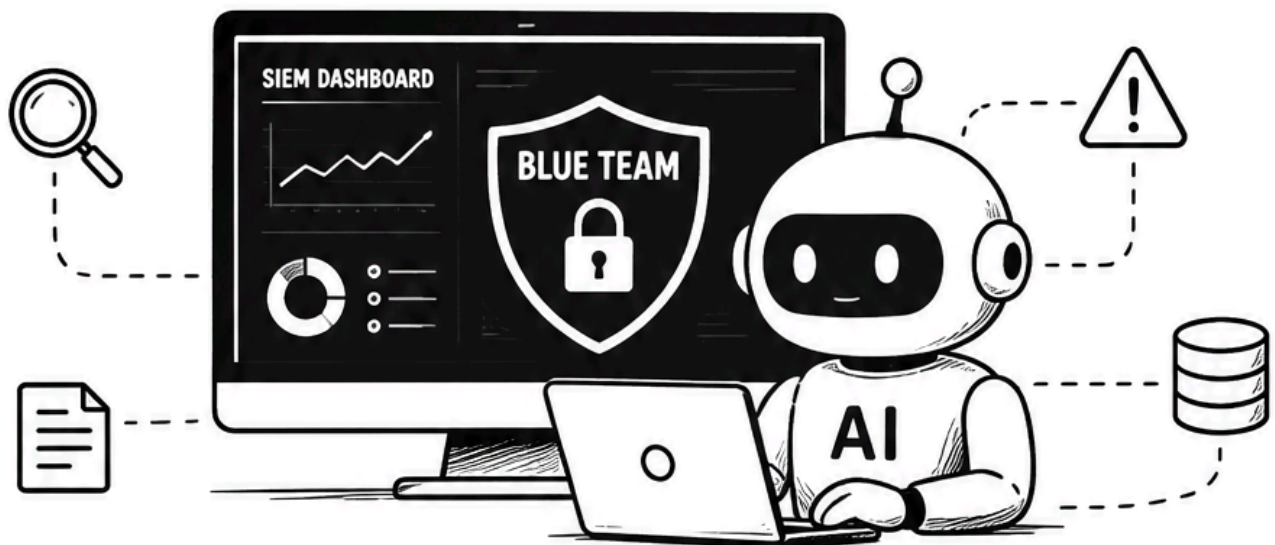


AI IS NOW YOUR SUPERPOWER IN THE SOC!

Work Faster. Detect Better. Stay Ahead.



In this guide, you will learn:

- ✓ Top AI tools every Blue Teamer should know
- ✓ What each tool does in simple words
- ✓ How to download and get started
- ✓ Real use cases in SOC environments
- ✓ How AI makes Blue Team work easier and smarter



AI won't replace Blue Teamers.

But Blue Teamers who use AI will replace those who don't.

WHY BLUE TEAMS USE AI TOOLS?



Blue Teamers protect systems, data, and users. They deal with huge amounts of alerts, logs, and reports. AI tools help them work faster, reduce noise, and focus on real threats.

WHAT AI HELPS WITH IN THE SOC



1. ANALYZE FASTER

AI reads logs and data in seconds.



2. REDUCE ALERT NOISE

AI filters out false alerts and shows what matters.



3. WRITE REPORTS

AI helps create summaries and reports quickly.



4. THREAT INTELLIGENCE

AI collects and summarizes threat data.



5. AUTOMATE TASKS

AI handles boring and repetitive work.



6. BETTER DECISIONS

AI gives insights to help analysts act fast.

WHERE AI HELPS MOST



SIEM Monitoring



Threat Hunting



Malware Analysis



Phishing Analysis



Incident Response



Remember:

AI is a helper, not a replacement.

It makes Blue Teamers faster, smarter, and more effective.

1. OLLAMA

Run AI models on your own computer.

? WHAT IS IT?

Ollama lets you run large language models (LLMs) locally.
Your data stays on your machine.
Private, secure, and offline-friendly.

🎯 WHY BLUE TEAMERS USE IT?



- Keep SOC data private
- Work offline
- Fast and lightweight
- Great for analysis, reports, and automation

⬇ HOW TO INSTALL (LINUX)

- 1 Open your terminal.

```
curl -fsSL https://ollama.com/install.sh | sh
```



- 2 Run the install command.

```
ollama --version
```



- 3 Verify installation.

```
# You should see the version number
```

🚀 HOW TO RUN A MODEL

- 1 Pull a model (example: llama3)

```
ollama pull llama3
```



- 2 Run the model

```
ollama run llama3
```

```
# Start chatting with AI
```



OLLAMA LOGO



>_ EXAMPLE (TERMINAL)

```
$ ollama run llama3
>>> Summarize the following alert in simple words:
Possible brute force login attempts detected
from IP 192.168.1.45 on multiple accounts.

llama3: This alert shows that someone is trying
many usernames and passwords to log in from the
same IP address. It could be a brute force attack.
The SOC analyst should check the IP and block it
if needed.
```

👥 USE CASES IN SOC



Log analysis
Ask AI to explain logs
in simple words.



Incident summary
Create quick summaries
of alerts.



Malware notes
Understand malware
behavior.



Report writing
Help draft SOC reports
faster.



PRO TIP

Use smaller models for daily tasks (like llama3, mistral).
Good balance of speed, privacy, and quality.

2. SHELLGPT

AI assistant for your terminal.

? WHAT IS IT?

ShellGPT is an AI tool that helps you generate and understand Linux commands using natural language. It works inside your terminal.



🎯 WHY BLUE TEAMERS USE IT?



- Quick Linux command help
- Faster log analysis
- Useful for investigations
- Saves time on daily tasks
- Works directly in terminal

📥 HOW TO INSTALL (LINUX)

- 1 Make sure Python is installed.

```
python3 --version
```

- 2 Install ShellGPT using pip.

```
pip install shell-gpt
```

- 3 Verify installation.

```
sgpt --version
```

```
john@soc-analyst:~$ sgpt --version
sgpt 1.6.6

john@soc-analyst:~$ sgpt --help
ShellGPT - AI assistant for your terminal
Usage: sgpt [options] <your question>
Example: sgpt "find large files in Linux"
```

>_ HOW TO USE

- 1 Type your question in simple English.
- 2 ShellGPT gives you the command.
- 3 Copy and run the command.

```
john@soc-analyst:~$ sgpt "find failed login attempts in Linux"
ShellGPT:
grep "Failed password" /var/log/auth.log | awk '{print $1}' |
sort | uniq -c | sort -nr
# This command shows IPs with failed login attempts.
```

📝 EXAMPLE USE CASES



Log Analysis
Find errors, failed logins, suspicious activities.



File Search
Find files, large files, modified files.



System Info
Check system info, running processes, users.



Network Info
Check open ports, connections, routes.

[>_] MORE EXAMPLES

QUESTION (YOU TYPE)	COMMAND (SHELLGPT GIVES)
show top 10 processes in Linux	<code>ps aux --sort=-%cpu head -n 10</code>
find files modified in last 7 days	<code>find / -type f -mtime -7 2>/dev/null</code>
check disk usage	<code>df -h</code>
find IP of a domain example.com	<code>dig +short example.com</code>



PRO TIP

ShellGPT does not run commands automatically. It only suggests commands. You stay in control.



NOTE

Always review commands before running, especially in production systems.

3. LOCALGPT

Chat with your own files using AI.

? WHAT IS IT?

LocalGPT lets you chat with your own documents, PDFs, and notes.

It keeps everything local and private.
No data is sent to the internet.



🎯 WHY BLUE TEAMERS USE IT?



- Search SOC reports quickly
- Summarize threat intel documents
- Understand malware reports
- Ask questions from your notes
- 100% private and offline

⬇ HOW TO INSTALL (LINUX)

1 Clone the repository.

```
git clone https://github.com/PromptEngineer/LocalGPT.git
```

2 Go to the directory.

```
cd LocalGPT
```

3 Install dependencies.

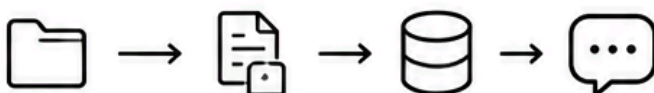
```
pip install -r requirements.txt
```

4 Run the setup.

```
python3 run_localGPT.py
```

```
user@linux:~$ cd LocalGPT
user@linux:~/LocalGPT$ python3 run_localGPT.py
LocalGPT is starting...
Loading documents...
LocalGPT is ready!
Open http://localhost:5111 in your browser.
(Default login: admin / admin)
```

⚙ HOW IT WORKS



1. Add Your Documents
2. Documents are split into chunks
3. Stored in Vector DB
4. Ask questions and get answers

📄 EXAMPLE QUESTION

Q: What are the steps for incident response in our playbook?

LocalGPT:

1. Identify the incident
2. Contain the threat
3. Eradicate the threat
4. Recover systems
5. Lessons learned

🛡 USE CASES IN SOC



Search SOC Reports

Find answers from old incident reports in seconds.



Malware Analysis Notes

Ask questions from malware analysis documents.



Playbooks & SOPs

Quickly find steps and procedures.



Threat Intelligence

Search threat reports and IOCs easily.



Internal Knowledge Base

Create your own private AI knowledge assistant.

📄 SUPPORTED FILE TYPES



PDF



TXT



DOCX



PPTX



Markdown



CSV



HTML



JSON

...and more



PRO TIP

Use LocalGPT with Ollama (local LLM) for fully offline and secure AI experience.



NOTE

Add only safe and company-approved documents.

3. LOCALGPT (EXAMPLE)

Live Example in Terminal

WHAT WE WILL DO?



We will load a SOC report (PDF) and ask LocalGPT a question. It will read the file and answer.

OUR TEST FILE



File Name: incident_report.pdf
Type: SOC Incident Report
Size: 2.1 MB

RUN LOCALGPT (TERMINAL)

1 Start LocalGPT

```
python3 run_localGPT.py
```

2 Open in Browser

```
http://localhost:5111
```

3 Login with default username and password

```
admin / admin
```

4 Upload your file

```
incident_report.pdf
```

5 Ask your question Get instant answer!

```
user@linux:~/LocalGPT$ python3 run_localGPT.py
LocalGPT is starting...
Loading documents...
LocalGPT is ready!
Open http://localhost:5111 in your browser.
(Default login: admin / admin)

-----
INFO:   Uvicorn running on http://0.0.0.0:5111
INFO:   Application startup complete.
INFO:   LocalGPT is running successfully.
```

EXAMPLE CHAT (TERMINAL)

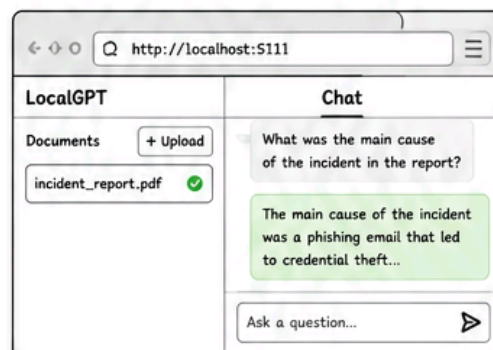
```
user@linux:~/LocalGPT$ localgpt chat
LocalGPT Chat Mode (type 'exit' to quit)

You: What was the main cause of the incident in the report?
LocalGPT:
The main cause of the incident was a phishing email that led to credential theft. The attacker used the stolen credentials to access the VPN and move inside the network.

You: What actions were taken?
LocalGPT:
1. User accounts were reset.
2. Malware was removed from affected systems.
3. Firewall rules were updated.
4. Users were trained on phishing awareness.

You: exit
Goodbye!
```

WHAT YOU SEE IN BROWSER



TIPS

- Split large files into smaller chunks for better results.
- Use clear and simple questions.
- Works best with good quality security documents.

COMMON QUESTIONS YOU CAN ASK

- Summarize this incident report.
- What are the indicators of compromise in this report?
- What is the timeline of the incident?
- What steps are recommended in this report?



USE CASE EXAMPLE



SOC Report
(PDF)



LocalGPT



Get Answers
Instantly



KEY TAKEAWAY:

LocalGPT helps Blue Teamers search, read, and understand security documents in seconds. It keeps everything local, private, and fast.

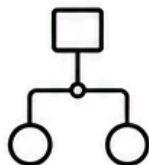


4. FLOWISE

Build AI workflows visually. No complex coding.

? WHAT IS IT?

Flowise is an open-source tool that lets you create AI workflows using a simple drag-and-drop interface.



🎯 WHY BLUE TEAMERS USE IT?



- Build SOC automation workflows
- Connect LLMs with your data
- Enrich alerts and IOC lookups
- No heavy coding required
- Fast prototyping and testing

⬇️ HOW TO INSTALL (LINUX)

1 Install Node.js (if not installed)

```
sudo apt update && sudo apt install nodejs npm
```

2 Install Flowise globally

```
npm install -g flowise
```

3 Verify installation

```
flowise -v
```

```
user@linux:~$ flowise -v
@flowise/cli/1.6.5 linux-x64 node-v18.16.0
user@linux:~$
user@linux:~$ flowise start
[INFO] Starting Flowise...
[INFO] Server running at http://localhost:3000
Open your browser and go to the above address.
Press Ctrl+C to stop.
```

🖥️ RUN FLOWISE

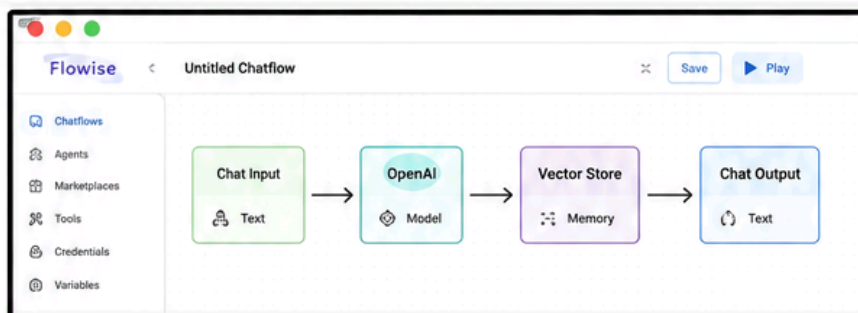
1 Start Flowise

```
flowise start
```

2 Open in browser

```
http://localhost:3000
```

3 Create your first workflow!



🔗 SIMPLE WORKFLOW EXAMPLE



This workflow takes an alert, extracts IoCs, checks threat intel, and gives a summary to the analyst.

🛡️ USE CASES IN SOC

- Alert Enrichment**
Add threat intel to alerts automatically.
- Phishing Analysis**
Analyze URLs, emails, attachments.
- Incident Summarization**
Summarize long alerts and reports.
- Automation**
Connect tools and reduce manual work.

>_ EXAMPLE TERMINAL OUTPUT

```
user@linux:~$ flowise start
[INFO] Starting Flowise...
[INFO] Server running at
http://localhost:3000
[INFO] Docs: https://docs.flowiseai.com
[INFO] Press Ctrl+C to stop
^C
user@linux:~$
```

📄 USEFUL LINKS

- Website: <https://flowiseai.com>
- GitHub: github.com/FlowiseAI/flowise
- Docs: docs.flowiseai.com



Flowise =
Visual AI workflows
for real-world SOC
automation.



PRO TIP

Use Flowise with LocalGPT or Ollama to build powerful offline AI workflows without sending any data to the internet.



4. FLOWISE (EXAMPLE)

Build a real workflow – Step by Step

WHAT WE WILL BUILD?

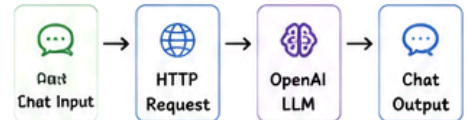
A simple SOC Assistant workflow that takes an analyst question, looks up threat intel (mock), and gives a clean answer.



NODES WE WILL USE

- Chat Input**
Accepts the analyst question.
- HTTP Request**
Gets threat intel from any API.
- OpenAI LLM**
Processes the data and generates an answer.
- Chat Output**
Shows the final answer.

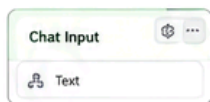
WORKFLOW OVERVIEW



- Analyst asks a question
- We call a threat intel API
- LLM understands the data
- Clean answer is returned to the analyst

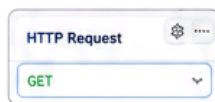
BUILDING THE WORKFLOW (IN FLOWISE UI)

1 Add Chat Input



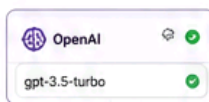
Drag "Chat Input" from the left panel.

2 Add HTTP Request



Use a threat intel API (e.g., AbuseIPDB, VirusTotal).

3 Add OpenAI LLM



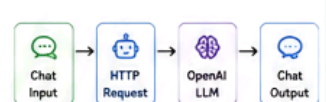
Connect your OpenAI API key in credentials.

4 Add Chat Output



Drag "Chat Output" to show the result.

5 Connect All



Connect nodes in order as shown above.

EXAMPLE: NODE CONFIGURATIONS

HTTP REQUEST NODE

Method: **GET**

URL:
`https://api.abuseipdb.com/api/v2/check`

Headers:

```

Key: Key
Value: YOUR_ABUSEIPDB_API_KEY
Key: Accept
Value: application/json
  
```

Query Params:
`ipAddress: {{input}}`

OPENAI LLM NODE

Model:
`gpt-3.5-turbo`

Temperature:
`0.2`

System Prompt:

```

You are a SOC analyst assistant.
Use the provided threat intel data
and answer clearly and concisely.
  
```

CHAT OUTPUT NODE

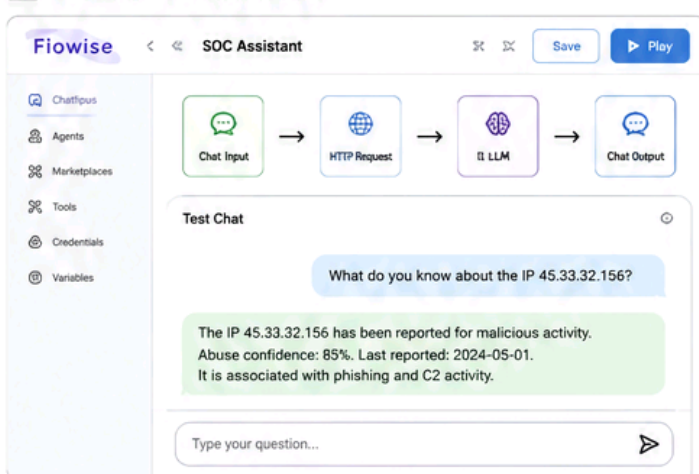
Output Text:

```

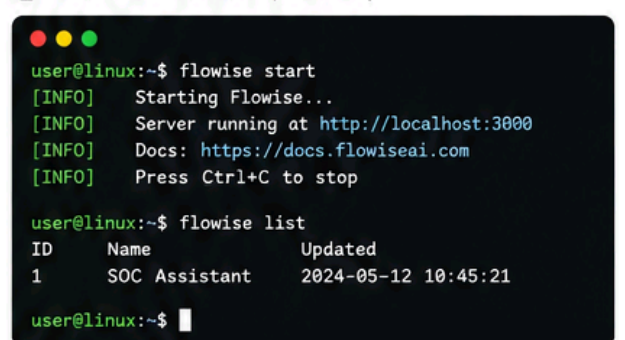
{{text}}
  
```

This will display the LLM answer to the analyst.

TESTING THE WORKFLOW



TERMINAL OUTPUT (FLOWISE)



TIP

You can export your workflow as JSON and share with your team.



KEY TAKEAWAY

Flowise lets you build powerful AI workflows visually. Perfect for automating SOC tasks without complex coding.



SECURITY REMINDER

Always use internal APIs and keep API keys in Flowise credentials (never hardcode).

5. LANGCHAIN

Build powerful AI apps with LLMs.

? WHAT IS IT?

LangChain is a framework for building LLM-powered applications using components like memory, chains, tools, and agents.



🎯 WHY BLUE TEAMERS USE IT?



- Build custom AI tools for SOC
- Connect LLMs to your data & tools
- Automate investigations
- Use memory for better context
- Integrate with many data sources

⬇️ INSTALL (PYTHON)

1 Create a virtual environment (optional)

```
python3 -m venv venv && source venv/bin/activate
```

2 Install LangChain

```
pip install -U langchain langchain-community
```

3 Verify Installation

```
python -c "import langchain; print(langchain.__version__)"
```

```
user@linux:~$ python -c "import langchain;
print(langchain.__version__)"
0.2.11
```

```
user@linux:~$
```

</> SIMPLE EXAMPLE (CHAT WITH YOUR DATA)

```
from langchain_community.document_loaders import TextLoader
from langchain.text_splitter import RecursiveCharacterTextSplitter
from langchain_community.vectorstores import FAISS
from langchain_community.embeddings import OllamaEmbeddings
from langchain.llms import Ollama
from langchain.chains import RetrievalQA

# 1. Load your document
loader = TextLoader("soc_playbook.txt")
docs = loader.load()

# 2. Split into chunks
splitter = RecursiveCharacterTextSplitter(chunk_size=1000, chunk_overlap=200)
splits = splitter.split_documents(docs)

# 3. Create embeddings + vector DB
embeddings = OllamaEmbeddings(model="nomic-embed-text")
vectorstore = FAISS.from_documents(splits, embeddings)

# 4. Create LLM
llm = Ollama(model="llama3")

# 5. Create QA chain
qa = RetrievalQA.from_chain_type(llm=llm, retriever=vectorstore.as_retriever())

# 6. Ask question
query = "What are the steps for incident response?"
result = qa.run(query)
print(result)
```

>_ RUN EXAMPLE (TERMINAL)

```
user@linux:~$ python langchain_example.py
> What are the steps for incident response?

LangChain:
1. Identify the incident
2. Contain the threat
3. Eradicate the threat
4. Recover systems
5. Lessons learned and documentation

user@linux:~$
```



WHAT HAPPENED?

1. Loaded your document (soc_playbook.txt)
2. Split into chunks
3. Created embeddings and stored in FAISS
4. Retrieved relevant chunks
5. LLM generated the answer

🔍 USE CASES IN SOC

Playbook Assistant
Ask questions from IR/DFIR playbooks.

Malware Analysis Helper
Get quick insights from malware reports.

Log Investigation
Summarize and analyze logs with AI.

Threat Intel Q&A
Ask questions from threat intel feeds.

🔗 KEY COMPONENTS

LLM The brain (e.g., Ollama, OpenAI)

Embeddings Convert text to vectors

Vector Store Store & search your data

Chain / Agent Combine steps & tools

Memory
Remember conversation context

🔌 COMMON INTEGRATIONS

📄 Data Sources: PDF, TXT, CSV, DBs, APIs

🔧 Tools: Bash, Python, Web Search

🗨️ LLMs: Ollama, OpenAI, Anthropic, etc.

🗄️ Vector Stores: FAISS, Chroma, Pinecone

🔗 Frameworks: FastAPI, Streamlit, Gradio



PRO TIP

Use LangChain with Ollama + Local data to build private, offline AI tools for your SOC.



NOTE

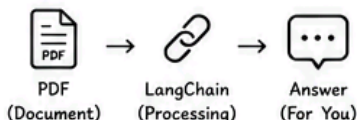
Ensure sensitive data never leaves your environment. Keep everything local when possible.

5. LANGCHAIN (EXAMPLE)

Build a Simple SOC Assistant with LangChain

WHAT WE WILL BUILD?

A simple SOC Assistant that can answer questions based on a security document (PDF). We will use LangChain to load the document, store it, and ask questions.



WHAT WE WILL USE?

- LangChain
- OpenAI (or any local LLM)
- FAISS (Vector Store)
- Python



TIP: You can use OpenAI API or a local LLM like Ollama.

HOW IT WORKS (EASY)?

1. Load the PDF document.
2. Split the text into small pieces.
3. Convert the text into numbers (embeddings).
4. Store the embeddings in a vector store.
5. Ask a question.
6. LangChain finds the best pieces and gives the answer.

EXAMPLE CODE (PYTHON)

```

1 from langchain_community.document_loaders import PyPDFLoader
  from langchain.text_splitter import RecursiveCharacterTextSplitter
  from langchain_community.vectorstores import FAISS
  from langchain_community.embeddings import OpenAIEmbeddings
  from langchain.llms import OpenAI
  from langchain.chains import RetrievalQA

2 # 1. Load your PDF
  doc = PyPDFLoader("soc_playbook.pdf")
  pages = doc.load()

3 # 2. Split into chunks
  splitter = RecursiveCharacterTextSplitter(chunk_size=1000, chunk_overlap=200)
  docs = splitter.split_documents(pages)

4 # 3. Create embeddings
  embeddings = OpenAIEmbeddings()

5 # 4. Store in FAISS vector store
  vectorstore = FAISS.from_documents(docs, embeddings)

6 # 5. Create LLM
  llm = OpenAI(temperature=0)

7 # 6. Create QA chain
  qa = RetrievalQA.from_chain_type(llm=llm, retriever=vectorstore.as_retriever())

8 # 7. Ask a question
  query = "What are the steps for incident response?"
  result = qa.run(query)
  print("Answer:\n", result)
  
```

Import the modules we need.

Load the PDF document.

Split the document into small pieces.

Convert text to numbers (embeddings).

Save the embeddings in FAISS (Vector Store).

Use OpenAI LLM (you can use Ollama too).

Create a question-answering chain.

Ask a question and get the answer!

RUN EXAMPLE (TERMINAL)

```

user@linux:~$ python langchain_example.py
Enter OpenAI API key: sk-*****

Question:What are the steps for incident response?

Answer:
1. Identify the incident
2. Contain the threat
3. Eradicate the threat
4. Recover systems
5. Lessons learned and documentation
  
```

EXPLANATION (SUPER SIMPLE)

- 1 We load a PDF (e.g., SOC playbook).
- 2 LangChain reads and splits the text.
- 3 It converts the text into numbers so the computer can understand.
- 4 It stores everything in a vector database (FAISS).
- 5 When you ask a question, it finds the most relevant text pieces.
- 6 The LLM uses those pieces to give you the answer!

CHANGE LLM (OPTIONAL)

You can use a local LLM with Ollama instead of OpenAI.

```

from langchain_community.llms import Ollama
llm = Ollama(model="llama3")
# Use the same code for QA chain
  
```

Make sure Ollama is running locally.

SAMPLE QUESTIONS YOU CAN ASK

- What are the incident response steps?
- How to handle a phishing incident?
- What are indicators of compromise (IOC)?
- What tools are recommended for malware analysis?
- How to report a security incident?



USE CASES IN SOC

- Summarize long reports
- Investigate alerts faster
- Get answers from playbooks
- Improve analyst productivity



KEY TAKEAWAY

LangChain helps you connect documents, data, and LLMs to build powerful AI assistants. It's flexible, private, and perfect for SOC workflows.

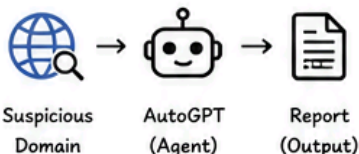


6. AUTOGPT (EXAMPLE)

A simple example – Investigate a phishing domain and create a report.

? WHAT WE WILL DO?

We will ask AutoGPT to investigate a suspicious domain and create a summary report.



🎯 OUR GOAL (EXAMPLE)

Goal:

"Investigate the domain evil-example.com, collect threat intel, and create a summary report."

- ✓ Find whois info
- ✓ Check reputation (VirusTotal)
- ✓ Look for related IOCs
- ✓ Create a summary report

⚙️ HOW AUTOGPT SOLVES IT

- 1 You give a GOAL.
- 2 AutoGPT breaks the goal into tasks.
Example: WHOIS lookup, reputation check, search for IOCs.
- 3 AutoGPT takes actions automatically.
It uses tools (browsing, API calls, file writing).
- 4 It learns from results and keeps going until the goal is done.
- 5 Finally, it creates a report for you!

📄 1. CREATE CONFIG (.env)

Create a file named .env and add your settings.

```

OPENAI_API_KEY=sk-*****
MEMORY_TYPE=local
MAX_TOKENS=2000
CONTINUOUS_MODE=True
BROWSE=True
  
```



Tip: Keep it simple for your first run!

📄 2. START AUTOGPT

- 1 Copy the example env (optional)
`cp .env.template .env`
- 2 Run AutoGPT
`python3 -m autogpt`
- 3 Enter your goal when asked

Goal:

"Investigate the domain evil-eample.com, collect threat intel, and create a summary report."

📄 TERMINAL (START)

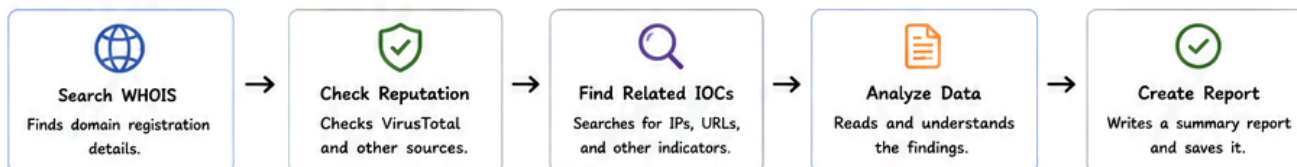
```

(venv) user@linux:~/AutoGPT$ python3 -m autogpt
Welcome to AutoGPT! Type your goal below.

Enter your goal: Investigate the domain
evil-example.com, collect threat intel,
and create a summary report.

Thinking...
PLAN:
- Search WHOIS for domain
- Check Passive DNS
- Check VirusTotal
- Search for related IOCs
- Write summary report
EXECUTING...
Goal in progress...
  
```

🤖 WHAT AUTOGPT DOES (EXAMPLE RUN)



GOOD TO KNOW: AutoGPT is autonomous. It decides the next step by itself based on the results it gets.

📄 TERMINAL OUTPUT (EXAMPLE)

```

Thinking... (step 7/25)
Action: WHOIS lookup for evil-example.com
Result: Domain created: 2023-08-15 | Registrar: NameCheap, Inc.

Thinking... (step 12/25)
Action: Checking VirusTotal
Result: 3 / 90 vendors flagged this domain as malicious.

Thinking... (step 18/25)
Action: Searching for related IOCs
Result: Found 2 related IPs and 4 related domains.

Thinking... (step 25/25)
Action: Writing report
Result: Report saved to /workspace/autogpt_workspace/report.md

GOAL COMPLETED! ✓
  
```

📄 REPORT PREVIEW (report.md)

```

# Phishing Domain Investigation Report

**Domain:** evil-example.com
**Date:** 2024-05-12

## Summary
The domain evil-example.com appears to be malicious.
It was flagged by 3/90 vendors on VirusTotal.

## Findings
- Registered: 2023-08-15 (NameCheap, Inc.)
- IP Address: 193.0.113.25
- VirusTotal: 3 / 90 detections
- Related Domains: evil-example.net, bad-login.com, ...

## Recommendations
- Block the domain and related IPs
- Monitor for related activity
- Investigate phishing reports
  
```



TIPS FOR BETTER RESULTS

- Give clear and specific goals.
- Start simple, then make it bigger.
- Review the plan before it runs.
- Check the reports and outputs.
- Use in a safe and isolated environment.



USE CASES IN SOC

- 🌐 Investigate phishing domains
- 🔍 Analyze suspicious files/URLs
- 📄 Collect threat intel automatically
- 📄 Create reports and summaries
- 👤 Monitor and track threat actors



IMPORTANT NOTE

AutoGPT can browse the internet and run commands. Always review its actions and outputs. You are in control.



Run locally. Protect your data and API keys.



KEY TAKEAWAY: AutoGPT is like a smart intern – you give it a goal, and it works for you until the job is done!

7. AI + SOC AUTOMATION

Work Smarter, Respond Faster, Stay Ahead.

? WHAT IS IT?

AI + SOC Automation uses AI tools, LLMs, and automation to handle repetitive tasks, speed up investigations, reduce alerts, and improve accuracy.



★ WHY USE IT?

- ✓ Reduce alert fatigue
- ✓ Investigate faster
- ✓ Automate repetitive tasks
- ✓ Improve threat hunting
- ✓ Consistent and accurate results
- ✓ Free up analysts for critical work



TIP: Automate the boring, focus on the important!

🎯 COMMON USE CASES



Alert Triage & Enrichment
Auto-enrich alerts with context (IP reputation, WHOIS, logs, etc.)



Incident Investigation
Summarize logs, timelines, and suggest next steps.



Threat Intelligence
Auto-collect and summarize threat intel from multiple sources.



Response Automation
Run playbooks, block IPs, create tickets, send notifications.

⚙️ HOW IT WORKS (HIGH LEVEL)



🔗 EXAMPLE WORKFLOW: ALERT TRIAGE AUTOMATION



Example:

A suspicious login alert comes in. AI enriches the IP, checks reputation, sees it's malicious, blocks it, and creates a ticket automatically.

🛠️ TOOLS YOU CAN USE

- SIEM: Splunk, Microsoft Sentinel, Elastic
- SOAR: Shuffle, TheHive, Cortex, Tines
- LLM/AI: OpenAI, GPT-4, Llama, Claude
- Automation: Python, PowerShell, n8n, Zapier
- Threat Intel: VirusTotal, AbuseIPDB, AlienVault OTX



TIP: Start small. Automate one task at a time!

</> EXAMPLE: ENRICH AN IP AND GET SUMMARY (PYTHON + OPENAI)

```
1 import requests, openai, json
2
3 ip = "45.33.32.156"
4
5 # 1. Get IP reputation from AbuseIPDB
6 api_key = "YOUR_ABUSEIPDB_API_KEY"
7 url = f"https://api.abuseipdb.com/api/v2/check"
8 headers = {"Key": api_key, "Accept": "application/json"}
9 params = {"ipAddress": ip}
10 resp = requests.get(url, headers=headers, params=params).json()
11 data = resp['data']
12
13 # 2. Create a summary with OpenAI
14 openai.api_key = "YOUR_OPENAI_API_KEY"
15 prompt = f"""
16 You are a SOC analyst.
17 IP: {ip}
18 AbuseIPDB Report:
19 {json.dumps(data, indent=2)}
20 Give a short summary, risk level and recommended action.
21 """
22 response = openai.ChatCompletion.create(
23     model="gpt-4o-mini",
24     messages=[{"role": "user", "content": prompt}]
25 )
26 print(response.choices[0].message.content)
```

> TERMINAL OUTPUT (EXAMPLE)

```
$ python ip_enrich_ai.py

--- AI SUMMARY ---
IP: 45.33.32.156

Risk Level: HIGH

Summary:
This IP has been reported for malicious activity multiple times.
Abuse confidence is high (85%). It is associated with phishing
and C2 activity. Last reported: 2024-05-01.

Recommended Action:
- Block the IP in firewall and EDR
- Monitor for related IoCs
- Create a ticket for incident response
```

🛡️ BEST PRACTICES

- Start with low-risk automations
- Always keep humans in the loop
- Test your playbooks thoroughly
- Log everything
- Continuously improve & tune

📈 BENEFITS

- Faster response
- Less manual work
- Better accuracy
- Happier analysts
- Stronger security posture

⚠️ CHALLENGES

- Data quality
- False positives
- Integration complexity
- Security & privacy
- Change management

🔒 IMPORTANT NOTE

Automation is powerful, but with great power comes responsibility. Always test, monitor and keep control.

You are the captain. AI is your co-pilot!



KEY TAKEAWAY: AI + SOC Automation helps you do more with less effort, respond faster, and focus on what matters most – protecting your organization!

READY TO LEVEL UP YOUR SOC WITH AI?

Join thousands of **Blue Teamers** learning, building, and winning together.

— LEARN. SHARE. AUTOMATE. DEFEND. —



PRACTICAL GUIDES

Step-by-step guides on AI tools for SOC professionals.



COMMUNITY

Learn and grow with a community of blue teamers.



REAL-WORLD USE CASES

See how AI is used in SOC's to detect, investigate & respond.



TOOLS & TEMPLATES

Playbooks, scripts, workflows, and more to save time.



STAY AHEAD

Stay updated with the latest AI trends in cybersecurity.



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